

Portable and low-cost 3D-printed visible-near infrared (vis-NIR) spectrometer using a linear charge-coupled device (CCD) sensor array

Bruno Vinicius Machado Castanho, Felipe Walter Dafico Pfrimer, Ricardo Schneider , Douglas José Coutinho

Federal University of Technology — Paraná (UTFPR), 85902-490 Toledo, PR, Brazil

ARTICLE INFO

Keywords:

Absorption spectroscopy
Instrumentation
Low-cost automation
Optical property
Embedded system
3D-printing

ABSTRACT

The spectrophotometer is one of the most valuable analytical instruments for quantitative spectral material analysis across various fields, measuring the intensity of light either emitted (luminescence) or transmitted (absorbance) by a sample. However, commercial portable spectrometers are typically expensive, and most low-cost alternatives have not shown performance comparable to commercial models. In this work, we present a low-cost and portable spectrometer utilizing a linear CCD sensor array (TCD1304DG) controlled by an STM32 board, enabling high-speed data acquisition and spectral resolution in the 400–1100 nm range comparable to that of commercial spectrometers. A user-friendly software application, developed in C# using the .NET framework, facilitates instrument calibration and operation. To validate the prototype, we measured the absorbance of various dye solutions and a thin organic semiconductor film, achieving results comparable to those obtained with a commercial portable spectrometer. This spectrometer can be easily assembled using commercially available components or printable parts with a standard 3D printer, making it suitable for research and educational applications.

1. Introduction

The rise of open-source electronic prototyping platforms, which feature user-friendly hardware and software, along with advancements in 3D printing from CAD models, has created significant opportunities developing creative and innovative product prototypes [1]. This is particularly well-suited to the need for low-cost scientific instrumentation, as laboratory equipment is typically costly, difficult to customize, and often incompatible with field research [2,3]. Recent examples include the development of a low-cost Raman spectrometer for educational purposes [4], a 3D printer-based microscope for long-term live cell imaging within a tissue culture incubator [5], a compact potentiostat/galvanostat for general electrochemistry measurement [6], a portable impedance analyzer for point-of-care diagnostic systems [7], and spectrophotometers [8,9], among others. In particular, absorption spectroscopy is a widely used method to determine the amount, composition, structure, and properties of a substance in many fields, including astronomy, chemistry, materials science, and physics.

Portable commercial spectrometers [10–13] are typically built using three main components: (i) an entrance slit, (ii) an aberration-corrected

concave holographic grating, which serves as both a focusing and dispersive element, and (iii) a CCD linear image sensor. The typical spectral range is between 350 and 1100 nm, primarily limited by the silicon responsivity of the linear CCD sensor. This spectrometer configuration allows for a very compact instrument enclosure and fast spectrum data acquisition, as all wavelengths are collected simultaneously, making it ideal for portable applications. However, these commercial spectrometers typically cost several thousand dollars, and when factoring in the light source—often similarly priced—the total cost can easily exceed \$5,000, making these devices inaccessible specially for low income countries.

Several prototype portable and compact spectrophotometers have been reported in the literature intended for applications in chemical sensing, education, pH sensing, plasma glucose detection, and honey purity [8,9,14–18]. However, none of these prototypes have demonstrated capabilities comparable to commercial devices, as most rely on smartphones or digital webcams as light sensors, CD-ROMs as diffraction gratings, and/or LEDs as light sources.

In this paper, we present a digital spectrometer built using a linear CCD sensor array as the light sensor, combined with a tungsten light

* Corresponding author.

E-mail address: douglascoutinho@utfpr.edu.br (D.J. Coutinho).

<https://doi.org/10.1016/j.measurement.2025.118265>

Received 6 December 2024; Received in revised form 17 April 2025; Accepted 24 June 2025

Available online 25 June 2025

0263-2241/© 2025 Elsevier Ltd. All rights reserved, including those for text and data mining, AI training, and similar technologies.

source to provide a spectral response in the Visible Near-Infrared (VIS-NIR) range (400–1100 nm). The sensor is controlled by an STM32 development board, enabling fast data acquisition on the millisecond time scale. Light diffraction is achieved with a transmission diffraction grating, and the spectrometer demonstrates a resolution comparable to that of commercial digital spectrometers. The housing and mechanical components of the prototype were fabricated using a 3D printer, ensuring a low-cost and customizable design. Open-source control software, developed in C# and the .NET framework is also provided for easy instrument control and calibration. The prototype was initially validated by comparing emissions from monochromatic LEDs, a white LED, and a tungsten lamp. Finally, the absorbance of different dye solutions (methyl red and methylene blue) and an organic semiconductor thin film (P3HT) was measured using both the prototype and a commercial spectrometer for comparison. The prototype demonstrated data quality similar to a commercial digital spectrometer while being low-cost, compact, and portable—features that make it especially suitable for teaching and field applications.

2. Materials and methods

The spectrometer developed here consists of two main 3D-printed parts: the lamp housing and the spectrometer box, where all the optical and electronic systems are assembled. This modular design allows for easy changes to the light source type, adaptation of an optical fiber to the spectrometer, and also enables fluorescence measurements with the help of a laser. It is important to mention that this spectrometer is intended to be open-source under the MIT License, and all files for 3D printing, PCB design, and software are available in a repository linked in the [Supplementary Material](#).

2.1. Lamp housing

A crucial step for good spectrum data acquisition is ensuring the light source's luminosity stability. To achieve this, we used a stable switched-mode power supply (SMPS) that converts AC 110/220 V to DC 12 V to power the light source and a generic 12 V fan for lamp cooling. The cooling is necessary because the halogen lamp generates significant heat, which could melt the nearby 3D-printed plastic parts. Additionally, a rectangular aluminum plate was glued to the inner wall of the lamp housing to aid in heat dissipation. To cover the visible and near-NIR spectral range (400–1100 nm) in this spectrometer, we used a standard 12 V G4 20 W tungsten halogen lamp as the light source. The luminosity level can be adjusted with a knob that controls the applied voltage to the lamp from 0 V to 12 V. This is done through an operational amplifier (LM358) and a MOSFET (BUZ71A), as shown in [Fig. 1a](#). Finally, the 3D-printed lamp housing, depicted in [Fig. 1b](#), also includes a cuvette and a thin slide groove, enabling absorbance measurements of solutions or thin-film samples. Images of the final assembly ([Fig. SM1](#)) as well as the complete list of bills of materials ([Table SM1](#)) are shown in the [Supplementary Material](#).

2.2. Spectrometer description

First, the incoming light passes through a single-slit formed by two trapezoidal blades typically found in utility knives ([Fig. 2](#)). The slit aperture was set to 100 μm by simply placing a razor blade in between the two trapezoidal blades. Next, a 52 mm convex lens is positioned at a distance $s = 75$ mm from the single-slit, forming an image at approximately $s' = 170$ mm, where a linear CCD is placed. Following this, the white light is dispersed into different wavelengths by a transmission diffraction grating with 500 grooves/mm, positioned 95 mm from the convex lens. The grating is oriented so that the incident light covers the maximum grating area, enhancing spectral resolution. The dispersed wavelengths are then focused onto different positions of the linear CCD image sensor (TCD1304DG), which consists of 3648 silicon photodiodes.

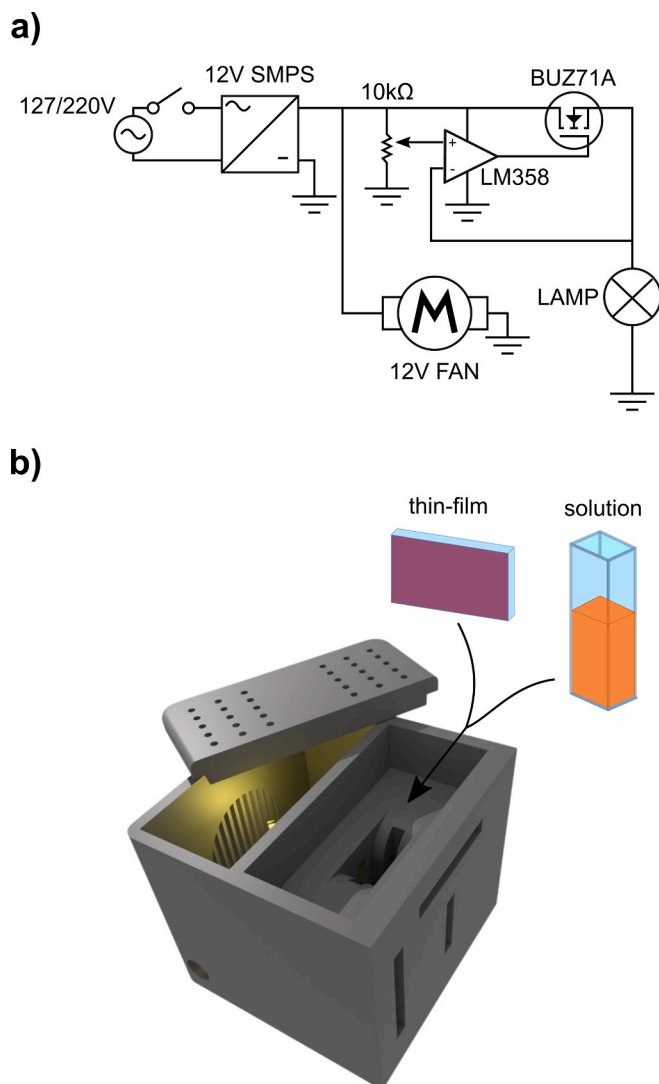


Fig. 1. a) lamp-housing powering circuit. A switched-mode power supply converts the input AC 110/220 V voltage to DC 12 V in which a potentiometer is used to control the lamp voltage thus adjusting its luminosity. b) 3D-printed lamp housing assembly, featuring compartments for absorbance measurements of solutions and thin-film samples.

These photodiodes absorb the light and convert it into a readable voltage signal. The CCD sensor is positioned 75 mm from the diffraction grating at a specific angle of 22° , calculated using the diffraction equation $d \sin \theta = \lambda$, where the distance d is the grating spacing (2 μm), and λ is the central wavelength of 750 nm. This configuration allows the sensor to capture the 400–1100 nm spectral range across its 3648 pixels.

The final resolution of the spectrometer is determined by three main factors: i) the number of pixels in the linear CCD, ii) the number of grooves N in the diffraction grating, and iii) the slit aperture. First, considering the high number of pixels covering the 400–1100 nm spectral range, the spectrometer would allow for a resolution of 0.2 nm per photodiode/pixel. For the diffraction grating, the resolution can be estimated from the full width at half maximum (FWHM) value using the relation $\Delta \theta = \lambda / Nd$, where $\Delta \theta$ is the half of FWHM, λ is the central wavelength of 750 nm, and N is the number of grooves illuminated on the diffraction grating (500/mm $\times$ 10 mm). The diffraction angle $\Delta \theta$ is then converted to wavelength using the diffraction equation, resulting in an FWHM of 0.3 nm. Lastly, the slit aperture is the most critical factor in determining the spectrometer's resolution. The narrower the slit aperture, the finer the incoming signal, and thus, the higher the resolution.

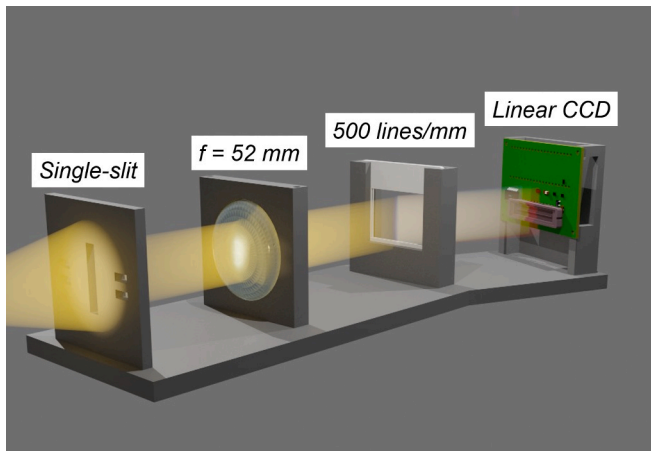


Fig. 2. 3D schematic view of the main components of the spectrometer. All supports for optical and electronic components were 3D-printed in specific fixed positions to allow for easy assembly. The distances and properties of the elements were chosen to maximize spectral resolution while keeping the prototype compact.

However, the aperture cannot be infinitely narrow because increased light diffraction and decreased light intensity would counteract resolution improvements. Therefore, the optimal slit width depends on the specific application for which the spectrometer is designed and can be adjusted accordingly. In this manuscript, the actual resolution of the spectrometer will be determined by measuring the emission spectra of LEDs and lamps, and comparing the results with those obtained from a commercial portable spectrometer.

2.3. Data acquisition and control

Data acquisition begins with light absorption by the silicon photodiodes that compose the linear CCD sensor. These photodiodes enable measurements in the visible and near-infrared spectra, ranging from 400 nm to approximately 1100 nm. Each photodiode is connected to a capacitor that stores the charge produced during an exposure time (integration time) set by the user, depending on the input light intensity. The voltage signals generated by all 3648-pixel elements are then processed by a shift register, which sends the output analog signal to a microcontroller. The microcontroller then converts this analog signal into a digital one enabling the data processing by the computer.

To properly configure the TCD1304DG linear sensor in the electric shutter function mode, three signal pins must be set: ICG (Integration Clear Gate), SH (Shift Gate), and ϕM (Master Clock), as illustrated in Fig. 3a. The master clock frequency is a reference digital signal sent by the microcontroller to the TCD1304DG sensor, and was set to the typical value of 2 MHz in this project, as recommended by the supplier's datasheet [19]. The ICG signal is responsible for clearing the charge stored in each capacitor, preparing the sensor for the next reading when the ICG signal is LOW. The ICG LOW time is set to 5 μs , with 1 μs being the minimum time stipulated by the datasheet. When ICG is HIGH, the photodiodes are enabled for measurement, and the total readout time is the sum of the ICG HIGH and ICG LOW times. Lastly, the SH pulse determines the integration time, controlling how long each pixel is exposed to light, and is the only parameter the user can set in the software interface with the lowest time limit of 10 μs . Additionally, the SH time also controls the charge transfer time of each photodiode to the shift register. All three signals must be well synchronized to ensure data integrity and prevent errors during the reading process.

Although the TCD1304DG is a highly sensitive linear CCD sensor, effective signal reading requires an external amplification circuit (Fig. 3b). The output signal (OS) from the TCD1304DG contains all the information about the light intensity captured by the sensor's

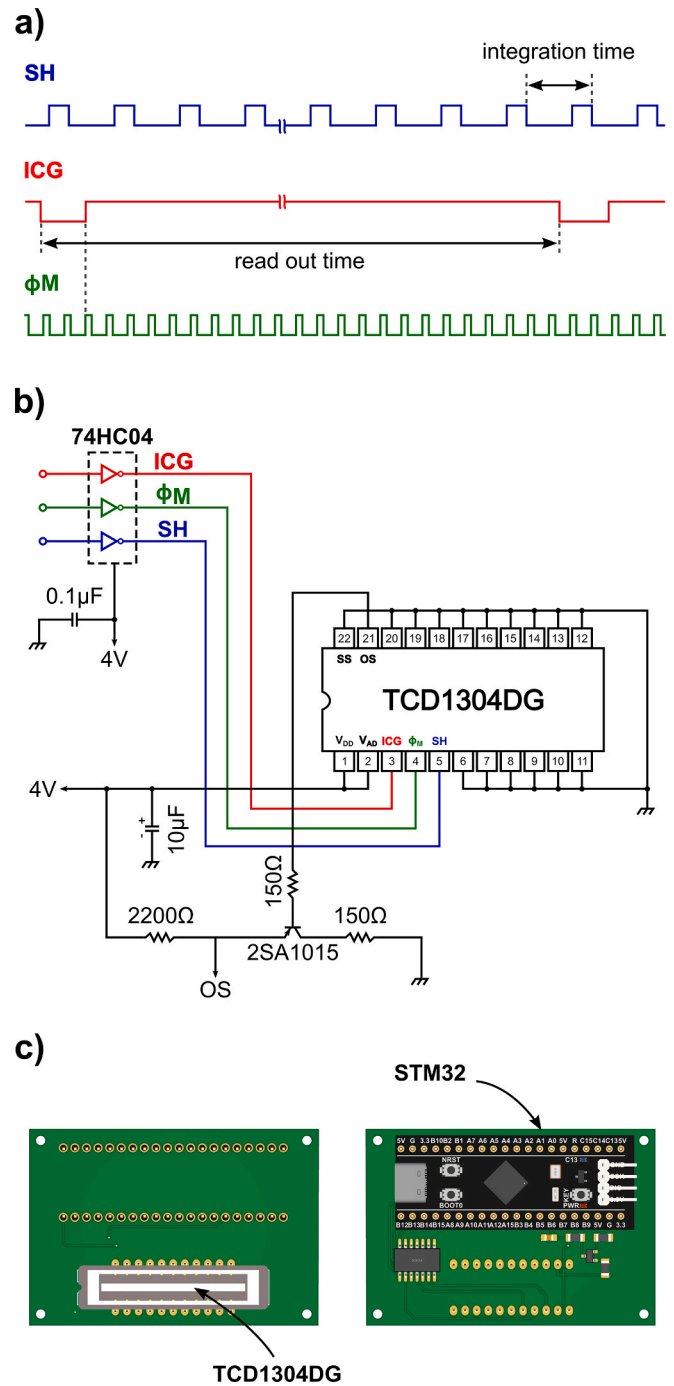


Fig. 3. a) Timing chart showing the three signals (ICG, SH, and ϕM) required to control the TCD1304DG linear CCD sensor. b) Control and signal amplification circuit. c) PCB integrating the linear CCD, electronic circuit, and microcontroller.

photosensitive elements. This signal, initially at very low voltage levels, must be amplified to ensure precise data processing in subsequent analog-to-digital (A/D) conversion steps. To achieve this, the OS is sent to a 2SA1015 PNP transistor, selected for its low saturation voltage, low noise, and high linearity characteristics. The base current, controlled by a resistor, sets the transistor's operating point and amplification gain. The amplified output voltage is then sent to the STM32F401CCU6 microcontroller, which performs the analog-to-digital conversion.

All signal control is managed by the STM32F401CCU6 microcontroller. This 32-bit microcontroller is based on the ARM Cortex-M4 architecture and offers high performance and energy efficiency—essential

for applications that require fast and accurate signal processing. The STM32 generates and synchronizes the ICG, SH, and ϕM signals, which are crucial for the TCD1304DG to operate in electronic shutter mode. Additionally, it performs A/D conversion of the amplified signal, digitizing it into 12 bits for real-time processing and analysis or later data processing.

A printed circuit board (PCB), shown in Fig. 3c, was developed to integrate all the electronics necessary for data acquisition and control. On this PCB, the linear CCD sensor and the microcontroller are directly connected, along with the amplification circuit described in Fig. 3b.

2.4. Software

A graphical interface (Fig. 4) was developed using C# and the .NET framework to manage communication between the PC and the STM32 microcontroller via the USB interface, utilizing the UART protocol. First, the COM port is selected, and by pressing the “Connect” button, communication is initiated, changing the status to “Connected”. The “Start” and “Stop” buttons begin and end data capture according to the measurement settings chosen by the user.

To set up the equipment for absorbance measurements, the CCD exposure time must first be configured. When the desired exposure time is selected, the software calculates the integration time, which represents the period during which the sensor accumulates charges from the incident light. The exposure time can be chosen from the dropdown menu shown in Fig. 4, with options ranging from one millisecond to several minutes. This approach enables the system to dynamically adapt to a wide range of lighting conditions. After setting the exposure time, the next step is to select an acquisition mode. In single mode, a single capture is performed upon the user’s command, while in continuous mode, measurements are taken continuously over time. Continuous mode is particularly useful for monitoring rapid changes in light intensity. In both modes, the software also provides an option to calculate the average of the measured intensities, helping to minimize the effects of momentary fluctuations and reading noise.

The final step for spectrometer initialization is to remove background noise by pressing the “Save Dark” button while blocking any incoming room light at the entrance slit. Next, the light source spectrum $I_0(\lambda)$ (baseline) is collected by turning on the spectrophotometer lamp and clicking “Save Base”. Once the baseline is stored, the sample can be

placed in the spectrometer, and by clicking “START”, real-time absorbance measurements can be obtained. Absorbance is calculated using the expression $A(\lambda) = \log\left(\frac{I_0(\lambda)}{I(\lambda)}\right)$, where $I(\lambda)$ represents the sample’s light intensity and $I_0(\lambda)$ is the baseline.

3. Results and discussion

The spectrometer’s performance was evaluated by acquiring emission spectra from known light sources, including monochromatic LEDs, a white-light LED, and a tungsten lamp, and comparing the results with those from a commercial digital spectrometer. The prototype was then used to measure the absorbance of a thin polymeric film and dye solutions.

3.1. Emission of light-sources

Before performing any absorbance measurement, it is essential to calibrate the spectrometer. This can be done using any known light source, provided it has well-defined, narrow peaks, such as lasers, LEDs, or gas-discharge lamps. In this study, we used two LEDs emitting at 456 nm and 640 nm, respectively, positioned 10 cm from the slit aperture. The calibration process involved acquiring the LED spectrum and entering the corresponding wavelengths into the Settings menu. The software then correlates the CCD pixel number with the respective wavelengths, completing the calibration process.

Fig. 5a and b show the LED emission spectra acquired by the prototype and the commercial digital spectrometer FLAME-T-XR1-ES from Ocean Insight. Fig. 5c and d display the emission spectra from a white LED and a 20 W tungsten halogen lamp. For these broad-spectrum light sources, a discrepancy between the two spectra is observed, caused by an interference effect due to multiple reflections of incoming light between the cover glass and the detector surface, as well as within the glass itself. This leads to the oscillating pattern seen in Fig. 5c, known as the etalon effect, commonly observed in many compact spectrometers [10].

Typical strategies to mitigate this interference include applying an anti-reflection coating to both sides of the detector window, removing the sensor window, or using a wedged cover glass. Alternatively, the total spectral response of the system can be compensated for by using the spectrum of a light source with known emissions, where each

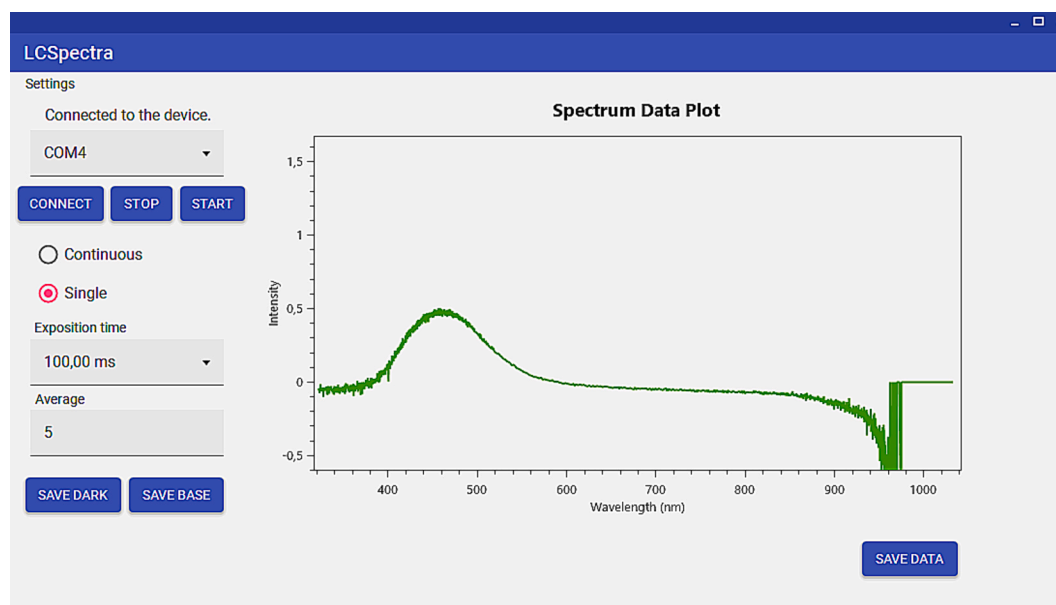


Fig. 4. Software interface for instrument connection and configuration, as well as data acquisition and plotting setup. The green graph displays an example absorption spectrum of a P3HT solution. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

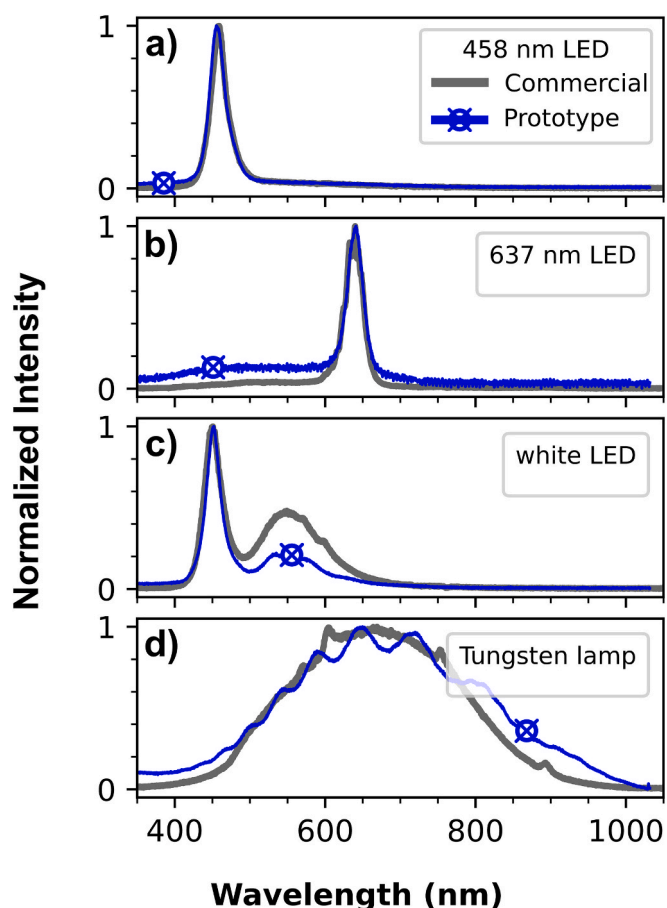


Fig. 5. Spectral emission of various light sources used for instrument calibration, measured by a commercial portable spectrometer (gray lines) and the prototype (blue lines). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

wavelength is multiplied by a correction factor, reducing the discrepancy caused by the etalon effect [20]. It is important to emphasize that this effect is critical only when measuring the absolute intensity of light sources or conducting color measurements. However, in the case of relative absorption spectroscopy, where the measurement is made relative to a reference spectrum, as in this project, the instrument's etalon effect is effectively canceled out.

3.2. Absorbance

To evaluate the spectrometer's performance, the absorbance of three different solutions and a thin organic semiconductor film was measured. The results from the prototype were compared with those obtained using a commercial digital spectrometer, the FLAME-T-XR1-ES from Ocean Insight. Fig. 6a–c present the absorbance of the following solutions: a) an aqueous solution of 3,7-bis(Dimethylamino)-phenothiazin-5-ium chloride, a commonly used dye and medication known as methylene blue (MB); b) an aqueous solution of 2-[[4-(Dimethylamino)phenyl]diazenyl] benzoic acid, also known as acid methyl red, typically used as a pH indicator; and c) a poly(3-hexylthiophene) (P3HT) solution at a concentration of 10 mg/mL in orthodichlorobenzene (ODCB). Fig. 6d shows the absorbance of a thin P3HT film deposited by spin coating at 1000 rpm for 180 s. P3HT is a common organic semiconductor used in the fabrication of organic electronic devices.

In all four tests, the absorbance peaks from both instruments aligned, demonstrating the effectiveness of the calibration process described in Section 3.1. Another important comparison is the absorbance magnitude, which also shows strong agreement between the prototype and the

commercial spectrometer. In terms of resolution, the prototype's capability was evaluated by detecting fine details, such as “shoulders” in the absorbance bands of MB at 620 nm and in the P3HT thin film at 580 nm. Specifically, MB exhibits two absorption maxima: the first band is assigned to the $n-\pi^*$ transition from amino groups, and the second is a weakly forbidden $n-\pi^*$ transition with charge transfer [21]. The shoulder observed in the P3HT spectrum at 580 nm indicates strong inter-chain $\pi-\pi$ stacking interactions, which result from improved molecular order in the conjugated polymeric thin film, enhancing its morphology for optoelectronic devices [22].

To verify the absorbance linearity according to the Beer-Lambert law, we performed absorbance-concentration measurements for both spectrometers using a P3HT solution monitored at 461 nm. The absorbance-concentration curve is shown in Fig. SM2, where both instruments exhibited a linear response with R^2 values of 0.98 (commercial) and 0.99 (prototype) within the concentration range of 8.5×10^{-4} to 1.5×10^{-4} mol/L. Additionally, we tested the upper and lower absorbance limits using an MB solution with concentrations ranging from 1×10^{-7} to 4×10^{-5} mol/L (Fig. SM3), where the lowest detectable absorbance was 0.02 and the highest was 2.00.

Finally, it is important to note that all measurements were performed without averaging or noise-reduction data processing. The increased noise observed at wavelengths below 450 nm and above 900 nm (not shown in Fig. 6) is due to the low emission of the tungsten lamp, as shown in Fig. 5, combined with the low responsivity of the silicon-based linear CCD sensor in this spectral range.

4. Conclusion

In this work, we present a portable digital spectrometer featuring a linear CCD sensor array as the light detector and a tungsten light source, designed for straightforward assembly with 3D-printed parts. The spectrometer interfaces with a computer and is controlled by an STM32 board, enabling rapid spectrum acquisition. The prototype demonstrated spectral performance in the Visible to Near-Infrared (VIS-NIR) range (400–1100 nm) with a resolution comparable to commercial spectrometers. Key parameters, such as the distances between the light source, entrance slit, converging lens, diffraction grating, and CCD sensor, were carefully optimized to achieve high spectral resolution, wide wavelength coverage, and adequate light intensity.

A user-friendly software application, developed in C# using the .NET framework, facilitates instrument calibration and operation. A straightforward calibration method is introduced, which can be performed with narrow-band light sources such as monochromatic LEDs, gas discharge lamps, or lasers. To validate the prototype, we measured the absorbance of various dye solutions and a thin organic semiconductor film, comparing the results with those obtained from a commercial portable spectrometer. The prototype demonstrated excellent agreement, delivering the best response observed so far to the best of our knowledge. Overall, this low-cost, easy-to-assemble spectrometer serves as a robust tool for portable absorption spectroscopy, suitable for scientific research and educational applications.

CRediT authorship contribution statement

Bruno Vinicius Machado Castanho: Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Felipe Walter Dafico Pfrimer:** Writing – review & editing, Validation, Methodology. **Ricardo Schneider:** Writing – review & editing, Methodology, Formal analysis, Conceptualization. **Douglas José Coutinho:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal

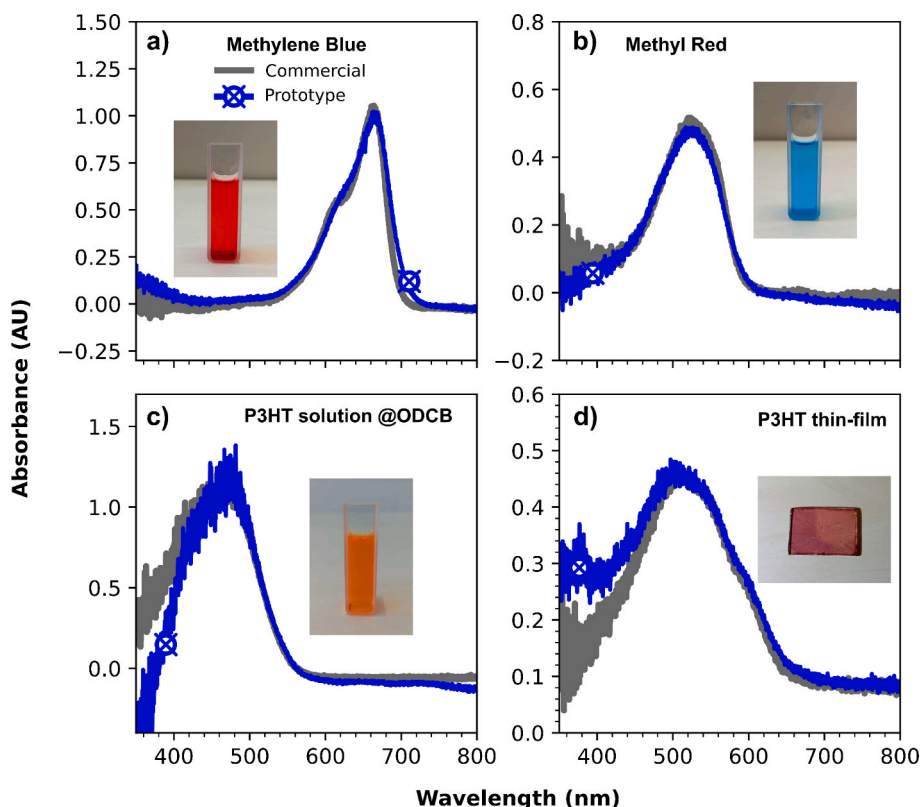


Fig. 6. Absorbance spectra of a) aqueous solutions of Methylene Blue and b) Methyl Red, c) a P3HT solution in orthodichlorobenzene (ODCB), and d) a P3HT thin film. Gray lines represent absorbance measured by a commercial portable spectrometer, while colored lines represent measurements by the prototype. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

relationships which may be considered as potential competing interests: Douglas José Coutinho reports financial support was provided by National Council for Scientific and Technological Development. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the Brazilian Foundations CNPq (Grant 14/50869-6), within the program of the National Institute for Science and Technology on Organic Electronics (INEO).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.measurement.2025.118265>.

Data availability

No data was used for the research described in the article.

References

- [1] A. Kantaros, O. Diegel, D. Piromalis, G. Tsaramiris, A.O. Khadidos, A.O. Khadidos, F.Q. Khan, S. Jan, 3D printing: making an innovative technology widely accessible through makerspaces and outsourced services, *Mater. Today Proc.* 49 (2022) 2712–2723, <https://doi.org/10.1016/j.matpr.2021.09.074>.
- [2] A. Tay, Lab kit on a budget: how cash-strapped research teams are getting creative, *Nature* (2024), <https://doi.org/10.1038/d41586-024-03366-6>.
- [3] A. Gatuszka, Z.M. Migaszewski, J. Namieśnik, Moving your laboratories to the field – Advantages and limitations of the use of field portable instruments in environmental sample analysis, *Environ. Res.* 140 (2015) 593–603, <https://doi.org/10.1016/j.envres.2015.05.017>.
- [4] N. Emmanuel, R.B. Nair, B. Abraham, K. Yoosaf, Fabricating a low-cost raman spectrometer to introduce students to spectroscopy basics and applied instrument design, *J. Chem. Educ.* 98 (2021) 2109–2116, <https://doi.org/10.1021/acs.jchemed.0c01028>.
- [5] G.O.T. Mercês, C. Kennedy, B. Lenoci, E.G. Reynaud, N. Burke, M. Pickering, The incubot: a 3D printer-based microscope for long-term live cell imaging within a tissue culture incubator, *HardwareX* 9 (2021) e00189, <https://doi.org/10.1016/j.ohx.2021.e00189>.
- [6] P. Irving, R. Cecil, M.Z. Yates, MYSTAT: a compact potentiostat/galvanostat for general electrochemistry measurements, *HardwareX* 9 (2021) e00163, <https://doi.org/10.1016/j.ohx.2020.e00163>.
- [7] L.A. Buscaglia, J.P. Carmo, O.N. Oliveira, Simple-Z: a low-cost portable impedance analyzer, *IEEE Sens. J.* 23 (2023) 26067–26074, <https://doi.org/10.1109/JSEN.2023.3312039>.
- [8] K. Laganovska, A. Zolotarjovs, M. Vázquez, K. Mc Donnell, J. Liepins, H. Ben-Yoav, V. Karitans, K. Smits, Portable low-cost open-source wireless spectrophotometer for fast and reliable measurements, *HardwareX* 7 (2020) e00108, <https://doi.org/10.1016/j.ohx.2020.e00108>.
- [9] B.J. Winters, N. Banfield, C. Dixon, A. Swensen, D. Holman, B. Fillbrown, 3D-printable and open-source modular smartphone visible spectrophotometer, *HardwareX* 10 (2021) e00232, <https://doi.org/10.1016/j.ohx.2021.e00232>.
- [10] Compact CCD Spectrometers, (n.d.). <https://www.thorlabs.com> (accessed November 8, 2024).
- [11] Spectrometers, Ocean Opt. (n.d.). <https://www.oceanoptics.com/spectrometers/> (accessed November 13, 2024).
- [12] Spectrometer System Home – StellarNet, Inc., (2023). <https://www.stellarnet.us/> (accessed November 13, 2024).
- [13] USB Spectrometer, Ossila (n.d.). <https://www.ossila.com/products/optical-spectrometer> (accessed November 13, 2024).
- [14] T.C. Muñoz-Hernández, E. González-Valencia, P. Torres, D.L. Aristizábal Ramírez, Low-Cost Spectrometer for Educational Applications using Mobile Devices, in: *Opt. Pura Apl.*, 2017: pp. 221–228. Doi: 10.7149/OPA.50.3.49035.
- [15] P. Hu, Y. Chen, S. Sonkusale, Low cost spectrometer accessory for cell phone based optical sensor, in: 2015 IEEE Virtual Conf. Appl. Commer. Sens. VCACS, 2015: pp. 1–5. Doi: 10.1109/VCACS.2015.7439566.
- [16] G. Likith, N.D. Jayram, B. Yaswanth, D. Sreekanth, J. Deny, M. Karuthapandi, S. V. Vishwa, Designing of low-cost spectrometer for sensor application, *J. Opt.* 50 (2021) 489–494, <https://doi.org/10.1007/s12596-021-00707-w>.
- [17] N. Chaianantakul, K. Wutthi, N. Kamput, N. Pramanpol, P. Janphuang, W. Pummara, K. Phimon, R. Phatthanakun, Development of mini-spectrophotometer for determination of plasma glucose, *Spectrochim. Acta. A.*

- Mol. Biomol Spectrosc 204 (2018) 670–676, <https://doi.org/10.1016/j.saa.2018.06.107>.
- [18] L.F. Bestari, P. Sarwanto, W.P. Syam, D. Harjunowibowo, A low-cost spectrometer to analyse the purity of honey, Phys. Educ. 57 (2021) 025004, <https://doi.org/10.1088/1361-6552/ac4083>.
- [19] alldatasheet.com, TCD1304DG Datasheet(PDF), (n.d.). <http://www.alldatasheet.com/datasheet-pdf/pdf/137658/TOSHIBA/TCD1304DG.html> (accessed November 13, 2024).
- [20] Spectral Response Correction: The Effect of Etaloning, AZoM (2017). <https://www.azom.com/article.aspx?ArticleID=14009> (accessed November 8, 2024).
- [21] Vibronic absorption spectrum and electronic properties of methylene blue in aqueous solution: TD-DFT study – ScienceDirect, (n.d.). <https://www.sciencedirect.com/science/article/pii/S016773222101093X> (accessed November 8, 2024).
- [22] J.Y. Na, B. Kang, D.H. Sin, K. Cho, Y.D. Park, Understanding solidification of polythiophene thin films during spin-coating: effects of spin-coating time and processing additives, Sci. Rep. 5 (2015) 13288, <https://doi.org/10.1038/srep13288>.